Corrected Winding Eddy-Current Harmonic Loss Factor for Transformers Subject to Nonsinusoidal Load Currents

A. Elmoudi, M. Lehtonen, H. Nordman

***Abstract*-- The assumption that the transformer windings eddy loss rise in proportion to the square of the harmonic order is reasonable for small conductors and low harmonics with errors on the conservative side only for a combination of large conductors and higher harmonics. This paper presents a corrected harmonic loss factor by taking into account the depth of penetration, which leads to more accurate estimation of the increased transformer winding losses under non-sinusoidal load currents.**

***Index Terms*—Harmonic loss factor, Transformers, IEEE C57-110-98**

I. Introduction

The significance of harmonics in power systems has increased substantially due to the use of solid state controlled loads and other high frequency producing devices. Many transformers installed to supply linear loads with sinusoidal currents have had their loads to be gradually replaced with non-linear loads that inject harmonic currents. These harmonic currents will increase losses and hence cause abnormal winding temperature rise which will decrease the expected lifetime. Such conditions require transformer derating to correspond to normal life expectancy or upgrading with a larger and more economical unit.

IEEE C57-110-98 standard [1], was developed to determine the capability of an existing transformer for nonsinusoidal currents. The eddy current loss is assumed to vary with the square of the rms current and the square of the frequency (harmonic order h). The term harmonic loss factor was developed to represent the effective rms heating as a result of the harmonic load current. Due to skin effect, the electromagnetic flux may not totally penetrate the strands in the winding at high frequencies. Thus, a power of 2 used in the calculation is on the conservative side [1] [2].

A harmonic loss factor was presented in [3] to predict the transformer capability to operate when supplying non-sinusoidal load currents. This paper provides an accurate harmonic loss factor that takes into account the depth of penetration. The presented corrected harmonic loss factor is compared with the harmonic order square standard factor [1] and the harmonic loss factor presented in [3].

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II. IEEE Recommended Practice C57.110-98

A. Review of Transformer Losses

Transformer losses are generally classified into no load or core losses and load losses

$$P_T = P_{NL} + P_{LL} \qquad W \tag{1}$$

where

P_{NL} is the core or no load loss.

P_{LL} is the load loss.

P_T is the total loss.

P_{NL} loss is the loss due to the voltage excitation of the core. Harmonic currents passing through the leakage impedance and the system impedance may slightly distort the transformer voltage. However, Experience has shown that the temperature rise of the core is not a limiting factor in determining the permissible current of non-sinusoidal loads.

P_{LL} is subdivided into P_{dc} and stray loss caused by electromagnetic field in the winding, core clamps, magnetic shields, enclosure or tank walls, etc. The P_{dc} loss is due to rms load current and dc resistance of the winding. The stray loss can be further divided into winding eddy loss and structural parts stray loss other than windings such as clamps, tanks or enclosure walls, etc.

$$P_{LL} = P_{dc} + P_{EC} + P_{OSL} \qquad W \tag{2}$$

Where

P_{dc} is the loss due to load current and dc resistance of the winding

P_{EC} is the winding eddy current loss.

P_{OSL} is the stray losses in clamps, tanks, etc.

The stray loss is determined by subtracting the P_{dc} loss from the load loss measured during the load loss test.

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$$P_{EC} + P_{OSL} = P_{LL} - P_{dc} \qquad W \tag{3}$$

The total specific winding loss at a certain spot can be calculated as:

$$P_W = P_{dc} + P_{EC} \qquad W/m^3 \tag{4}$$

The P_{dc} loss is

$$P_{dc} = \rho J^2 \tag{5}$$

Where

ρ is the conductor resistivity.

J is the rms current density.

Consider the conductor of Fig.1 is immersed in alternating magnetic field

$$B(t) = \sqrt{2}\ B\ \sin(h\omega t) \tag{6}$$

where

B is the rms value of the magnetic flux density.

f is the fundamental frequency 50 or 60 Hz.

h is the harmonic order

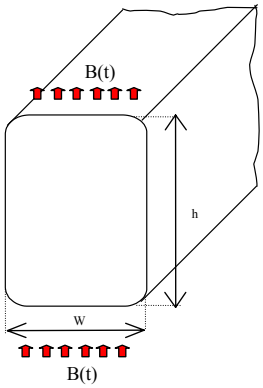


Fig.1. A conductor strand immersed in a magnetic field perpendicular to its width w

The contribution of eddy current losses caused by the magnetic field is obtained using [4]

$$P_{EC} = \frac{\pi^2 f^2 T^2 B^2}{3\rho} \qquad W/m^3 \tag{7}$$

Where

T is the conductor thickness (height h or width w) perpendicular to the direction of B.

The winding loss normalized with the rated P_{dc} loss can be expressed as

$$\frac{P_W}{P_{dc-R}} = \frac{P_{dc}}{P_{dc-R}} + \frac{P_{EC}}{P_{dc-R}} \tag{8}$$

Where the rated P_{dc} loss is

$$P_{dc-R} = \rho J_R^2 \tag{9}$$

where

J_R is the rms rated current density

When the load current is periodic, but non-sinusoidal, its rms value is

$$I = \sqrt{\sum_{h=1}^{h=h_{\max}} I_h^2} = A_c \sqrt{\sum_{h=1}^{h=h_{\max}} J_h^2} \tag{10}$$

Where

I_h is the rms current harmonic of order h,

J_h is the rms current density.

A_c is the conductor cross-sectional area.

h is the harmonic order.

$h_{\max}$ is the highest harmonic order to be considered.

The first term of (8) becomes

$$\frac{P_{dc}}{P_{dc-R}} = \frac{\rho J^2}{\rho J_R^2} = \left(\frac{I}{I_R}\right)^2 = \frac{\sum_{h=1}^{h=h_{\max}} I_h^2}{I_R^2} \tag{11}$$

The leakage magnetic field B, is directly proportional to the load current I,

$$B = K_B I \tag{12}$$

When the transformer load current is equal to the rated current, the eddy current loss is

$$P_{EC-R} = K I_R^2 = K A_c J_R^2 \tag{13}$$

Where

$$K = \frac{\pi^2 f^2 T^2 K_B^2}{3\rho}$$

The eddy current loss due to non sinusoidal load current becomes

$$P_{EC} = K \sum_{h=1}^{h=h_{\max}} h^2 I_h^2 = K A_c \sum_{h=1}^{h=h_{\max}} h^2 J_h^2 \tag{14}$$

From (13) and (14) we obtain:

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$$P_{EC} = KA_c J_R \sum_{h=1}^{h=h_{\max}} h^2 \left(\frac{J_h}{J_R}\right)^2 = P_{EC-R} \sum_{h=1}^{h=h_{\max}} h^2 \left(\frac{I_h}{I_R}\right)^2 \quad (15)$$

Hence, the second term of (8)

$$\frac{P_{EC}}{P_{dc-R}} = \frac{P_{EC-R}}{P_{dc-R}} \sum_{h=1}^{h=h_{\max}} h^2 \left(\frac{I_h}{I_R}\right)^2 \quad (16)$$

Equation (16) was derived based on the assumption that the measured application currents are taken at rated currents of the transformer.

Since this is seldom encountered in the field, a new term is defined based on the winding eddy losses at the measured current and power frequency.

Substituting (11) and (16) in (8) gives

$$\frac{P_W}{P_{dc-R}} = \frac{\sum_{h=1}^{h=h_{\max}} I_h^2}{I_R^2} \left[1 + \frac{P_{EC-R}}{P_{dc-R}} \frac{\sum_{h=1}^{h=h_{\max}} h^2 I_h^2}{\sum_{h=1}^{h=h_{\max}} I_h^2} \right] \quad (17)$$

or

$$P_W(pu) = I^2(pu)\left[1 + P_{EC-R}(pu) F_{HL}\right] \quad (18)$$

where

$I^2(pu)$ is the normalized current squared.

$P_{EC-R}(pu)$ is the normalized eddy current loss under rated conditions.

F_{HL} is the harmonic loss factor.

The harmonic loss factor is normalized to either the fundamental or the rms current

$$F_{HL} = \frac{\sum_{h=1}^{h=h_{\max}} h^2 I_h^2}{\sum_{h=1}^{h=h_{\max}} I_h^2} = \frac{\sum_{h=1}^{h=h_{\max}} h^2 \left(\frac{I_h}{I}\right)^2}{\sum_{h=1}^{h=h_{\max}} \left(\frac{I_h}{I}\right)^2} \quad (19)$$

The harmonic loss factor is a key indicator of the current harmonic impact on the winding eddy losses. Under rated sinusoidal current $I(pu) = 1,\ F_{HL} = 1$ and the hot spot specific power loss is

$$P_W(pu) = 1 + P_{EC-R}(pu) \quad (20)$$

B. Evaluating Load Capability of Existing Transformers

The recommended method in [1] for dry type transformers is based on the condition that the pu value of the non-sinusoidal current will cause the same hot spot losses as the rated sinusoidal current. This can be expressed as

$$I_{\max}(pu) = \sqrt{\frac{P_{W-R}(pu)}{1 + F_{HL} \times P_{EC}(pu)}} \quad (21)$$

This assumes that the normal life of the unit will be maintained.

III. Transformer winding Losses Computation using FEM

Two-dimensional finite element method is the most commonly used method to compute eddy losses. The analysis of the winding eddy loss by 3-D FEM will be more accurate, however the computation cost and the complexity increase and in many cases a reduction to a 2-D is possible without significant loss of accuracy.

For windings in which the conductors are small strands i.e. smaller than the depth of penetration, we can model the region as non-conducting region; find the resistance limited eddy current losses. The field solution gives the familiar leakage flux where the flux in the windings flows axially up through the coils and then bends radially across the windings. The leakage flux has its greatest concentration on the interface between the two windings and decreases, as progress is away from the gap between the windings. The inner LV winding typically has a higher attraction of the leakage flux due to the high permeance of the core. The HV winding divides its leakage flux with part being attached to the core and the reminder attracted to the core clamps and other structural parts.

The critical point in transformers is usually in the upper end of the windings where the conductors are exposed to an inclined magnetic field with two components, an axial component and a radial component, and the eddy current losses are the contribution of these two components. Using local flux density from FEM solution the eddy loss density can be calculated. In order to calculate the winding losses the P_{dc} loss must be added to the eddy loss [6].

IV. corrected harmonic loss Factor

At high frequencies conductor skin effect or the limited magnetic flux penetration has to be taken into account. A corrected harmonic loss factor is suggested based on the following equation [5].

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$$P_{EC} = \frac{\pi^2 f^2 T^2 B^2}{3\rho} F(\xi) \tag{22}$$

Where, $F(\xi) = \frac{3}{\xi} \frac{\sinh \xi - \sin \xi}{\cosh \xi - \cos \xi}$

$\xi = \frac{T}{\delta}$ is strand dimension related to skin depth.

$\delta_R = \sqrt{\rho / \mu \pi f}$ is the depth of penetration at power frequency 50 or 60 Hz.

$\delta = \frac{\delta_R}{\sqrt{h}}$ is the depth of penetration at harmonic frequency.

The function F(ξ) is as shown in Fig. 2. For small values of ξ, F(ξ) ≈ 1 and the loss at low frequencies is proportional to square of the frequency and the square of the conductor thickness.

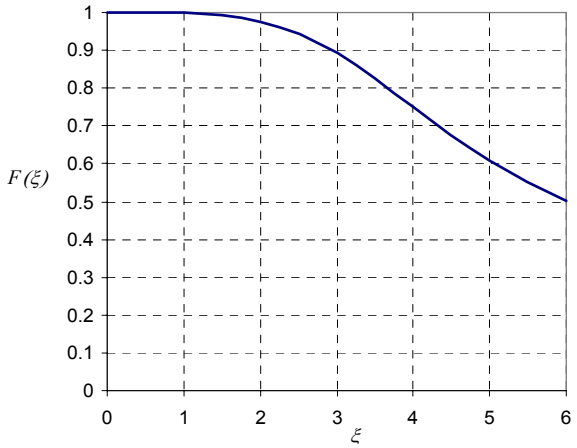


Fig. 2. The Function F(ξ)

For Copper conductor at 60 Hz and 75°C $\delta_R \approx 9.4$ mm and for Aluminum $\delta_R \approx 12.1$ mm.

The strand dimension related to skin depth for a harmonic order h is

$$\xi_h = \frac{T}{\delta_R / \sqrt{h}} = \xi_R \sqrt{h} \tag{23}$$

writing (13) as a function of the current

$$P_{EC} = KF(\xi) h^2 I^2 \tag{24}$$

when the current is non-sinusoidal

$$P_{EC} = K \sum_{h=1}^{h=h_{\max}} F(\xi)\ h^2 I_h^2 \tag{25}$$

Normalizing the eddy current loss produced by non-sinusoidal load current to the eddy current losses at rated condition a corrected harmonic loss factor is achieved.

$$\frac{P_{EC}}{P_{EC-R}} = \frac{\sum_{h=1}^{h=h_{\max}} F(\xi_h)\ h^2 I_h^2}{F(\xi_R)\ I_R^2} \tag{26}$$

or

$$\frac{P_{EC}}{P_{dc-R}} = \frac{P_{EC-R}}{P_{dc-R}} \frac{\sum_{h=1}^{h=h_{\max}} F(\xi_h)\ h^2 I_h^2}{F(\xi_R)\ I_R^2} \tag{27}$$

Substituting (11) and (27) in (8) gives:

$$\frac{P_W}{P_{dc-R}} = \frac{\sum_{h=1}^{h=h_{\max}} I_h^2}{I_R^2} \left[1 + \frac{P_{EC-R}}{P_{dc-R}} \frac{\sum_{h=1}^{h=h_{\max}} \frac{F(\xi_h)}{F(\xi_R)} h^2 I_h^2}{\sum_{h=1}^{h=h_{\max}} I_h^2} \right] \tag{28}$$

Equation (29) leads to a corrected loss factor

$$F_{HL} = \frac{\sum_{h=1}^{h=h_{\max}} \frac{F(\xi_h)}{F(\xi_R)} h^2 I_h^2}{\sum_{h=1}^{h=h_{\max}} I_h^2} = \frac{\sum_{h=1}^{h=h_{\max}} \frac{F(\xi_h)}{F(\xi_R)} h^2 \frac{I_h^2}{I^2}}{\sum_{h=1}^{h=h_{\max}} \frac{I_h^2}{I^2}} \tag{29}$$

The importance of the above corrected factor is demonstrated in Fig.3 where the graphs are presented as a function of the harmonic order, for different copper conductor thicknesses. The graph of the square harmonic order h^2 is added for comparison. For small conductors or at low harmonics the corrected harmonic loss factor approaches h^2 i.e. (28) becomes (17).

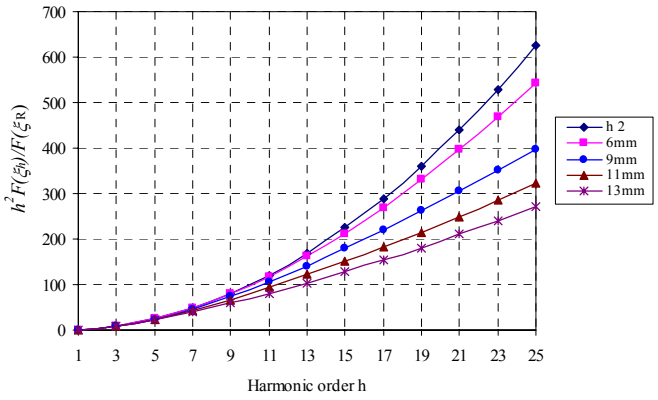


Fig. 3. The function $h^2F(\xi_h)/F(\xi_R)$ vs. harmonic order h for different rectangular copper conductor thickness.

It can be seen that for small conductors skin effect is insignificant and the assumed h^2 harmonic loss factor predicts the losses accurately. The assumed loss factor predicts losses which are higher than the actual case for large conductors and higher harmonics due to the ignored conductor penetration

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depth. The actual flux impinging on the conductor will be less due to skin effect.

The distribution of the magnetic flux density and the eddy current density in cross-section of the conductor are shown in Fig.4

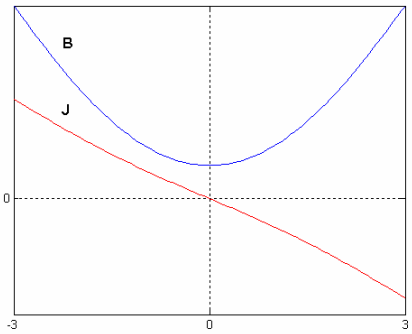


Fig. 4. Distribution of magnetic flux density B and eddy current J in the cross-section of a conductor immersed in a magnetic field.

The dimension of conductor sides used in transformers may be found in a wide range. The corrected harmonic loss factor presented can be used to count for both axial and radial eddy current.

V. Example Calculation for Dry Type Transformer

An example in [1] is reworked to compare the presented corrected harmonic loss factor with the harmonic order square and the harmonic loss factor suggested in [3]: it is a standard transformer with 1200 rated current and P_{EC-R} = 0.15. The secondary winding is assumed to consist of copper strands of 3.65×11 mm, given the non-sinusoidal load current with the harmonic distortion as shown in Fig.5.

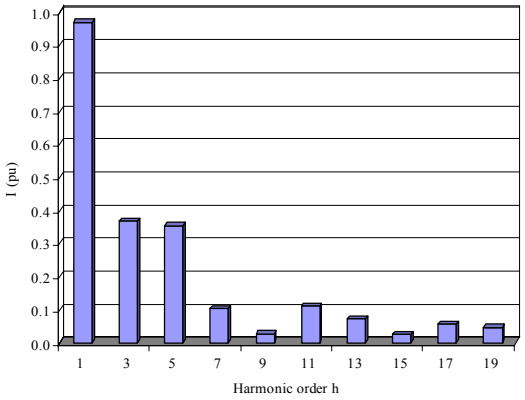


Fig. 5. The harmonic current spectrum load.

Harmonic Order Square

The normalized current spectrum and the data required for (19) is as in table I.

TABLE I
NORMALIZED CURRENT SPECTRUM AND CALCULATED DATA FOR HARMONIC LOSS FACTOR F_{HL}

h	(I_h/I)	$(I_h/I)^2$	h^2	$h^2 (I_h/I)^2$
1	1.0	1.000000	1	1.0000
5	0.2330	0.054289	25	1.3572
7	0.108	0.011664	49	0.5715
11	0.042	0.001764	121	0.2134
13	0.027	0.000729	169	0.1232
17	0.013	0.000169	289	0.0488
19	0.008	0.000064	361	0.0231
		1.0687		3.337

The harmonic loss factor F_{HL} is

$$F_{HL} = \frac{3.337}{1.0687} = 3.129$$

The total specific normalized losses are

$$P_W(pu) = 1.0687\left[1 + 0.15 \times 3.123\right] = 1.569 \ \ pu$$

The value of the maximum non-sinusoidal current yielding the same winding eddy losses as sinusoidal current is

$$I_{\max}(pu) = \sqrt{\frac{1.15}{1 + 0.15 \times 3.129}} = 0.885 \ \ pu$$

or

$$I_{\max} = 0.885 \times 1200 = 1062 \ \ A$$

Corrected Harmonic Loss Factor

The normalized current spectrum and the data required for (29) is as in table II.

TABLE II
NORMALIZED CURRENT SPECTRUM AND THE CALCULATED DATA FOR THE CORRECTED HARMONIC LOSS FACTOR F_{HL}

h	(I_h/I)	$(I_h/I)^2$	$h^2F(\xi_h)/F(\xi_R)$	$(h^2F(\xi_h)/(\xi_R))\ (I_h/I)^2$
1	1.000	1	1	1.000
5	0.233	0.054289	23.39	1.270
7	0.108	0.011664	43.27	0.505
11	0.042	0.001764	93.29	0.165
13	0.027	0.000729	121.59	0.089
17	0.013	0.000169	182.89	0.031
19	0.008	0.000064	215.7	0.014
		1.0687		3.072

The harmonic loss factor F_{HL} is

$$F_{HL} = \frac{3.072}{1.0687} = 2.875 \ \ pu$$

The total specific normalized losses are

$$P_W(pu) = 1.0687\left[1 + 0.15 \times 2.875\right] = 1.529 \ \ pu$$

The value of the maximum non-sinusoidal current yielding the same winding eddy losses as sinusoidal current is

$$I_{\max}(pu) = \sqrt{\frac{1.15}{1 + 0.15 \times 2.875}} = 0.9 \ \ pu$$

$$I_{\max} = 0.9 \times 1200 = 1075.65 \ \ A$$

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As in Table III the presented loss factor taking into account skin effect allow 1.32% increase only compared to the h^2-rule, due to the low harmonics spectrum, while the harmonic loss factor presented in [3], allow 8.6 % increase in the current.

TABLE III
COMPARISON OF THE MAX. ALLOWED NON-SINUSOIDAL LOAD CURRENT FOR DIFFERENT HARMONIC LOSS FACTORS F_{HL}

Harmonic Factor	F_{HL}	$I_{max}(A)$
$h^2 (I_h/I)^2$	3.129	1062
$(h^2F(\xi_h)/(\xi_R))(I_h/I)^2$	2.875	1075
$h(\xi_h)/F(\xi_R)(I_h/I)^2$	1.683	1157

VI. Conclusion

The assumption that the eddy current loss in transformer windings is proportional to the square of the frequency is reasonable for transformers with small conductors and subject to low harmonics. For a combination of large conductors and high harmonics such assumption leads to a conservative result. It predicts losses which are higher than the actual case due to the ignored penetration depth. The actual flux impinging on the conductor will be less due to skin effect.

A corrected harmonic loss factor which takes into account the depth of penetration can be applied to determine more accurately the capability of transformers subject to non-sinusoidal load currents.

VII. References

[1] *IEEE Recommended practice for Establishing Transformer capability when Supplying Nonsinusoidal load Currents*, IEEE Standard C57.110-1998, Jul. 98.

[2] S. P. Kennedy, C. L. Ivey, "Application, Design and Rating of Transformers Containing Harmonic Currents," *IEEE Pulp and paper industry conference*, 18-22 Jun. 90, pp.19-31.

[3] S. N Makarov, A. E. Emanual "Corrected Harmonic Loss Factor For Transformers supplying Non-sinusoidal Load current," Proc. of the 9th International conference on Harmonics and Power Quality, vol. 1, Oct. 2000, pp.87-90.

[4] Lates L. V., Electromagnetic calculation of Transformers and Reactors: Moscow ENERGY, 1981 (In Russian).

[5] J. Laminar and M. Stafl, *Eddy Currents*. London: Iliffe, 1966, p. 33.

[6] A. Elmoudi, M. Lehtonen "Eddy losses Calculation in Transformer Windings using FEM," Proc. of Rega Technical University International conference, Oct.. 9-11 2002, pp. 6.

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